

Testing Instructions Report - SafeMother

TABLE OF CONTENTS

1. Important Note.....	2
2. Running Unit Tests.....	2
2.1 Test Commands.....	2
3. Testing Environment Configuration.....	3
3.1 Framework & Runtime.....	3
3.2 Test Structure.....	3
3.3 Dependencies.....	4

1. Important Note

The unit tests reside in the **feature/unit-testing** branch and are **not merged** into the **development** or **main** branches.

To access the tests, switch to the branch (**feature/unit-testing**) using the following command:

git checkout feature/unit-testing

2. Running Unit Tests

All tests are located within the *server/* directory. Ensure you navigate to this directory before executing any test commands:

cd server

2.1 Test Commands

Command	Description
npm test	Runs all unit tests once
npm run test:watch	Runs tests in watch mode (automatically re-runs on file changes)
npm run test:coverage	Runs tests and generates a coverage report

3. Testing Environment Configuration

3.1 Framework & Runtime

Item	Value
Test Framework	Jest ^30.3.0
Test Environment	Node
ES Module Support	NODE_OPTIONS=--experimental-vm-modules (configured via cross-env)

3.2 Test Structure

The project follows a structured approach for organizing unit tests as shown below:

```

server/
  tests/
    setup/
      jest.setup.js          # Global afterEach hook to clear all mocks
    unit/
      controllers/           # Controller-level unit tests
        AppointmentController.test.js
        AuthController.test.js
        MessageController.test.js
        PregnancyController.test.js
        TipController.test.js
        UserController.test.js
      services/              # Service-level unit tests
        AppointmentService.test.js
        AuthService.test.js
        GeminiService.test.js
        MessageService.test.js
        PregnancyService.test.js
        TipService.test.js
        UserService.test.js

```

3.3 Dependencies

The following development dependencies are required for running the tests:

```
"devDependencies": {  
  "cross-env": "^10.1.0",  
  "jest": "^30.3.0"  
}
```

Before running the tests, ensure all dependencies are installed:

```
npm install
```